



CS 641

Midsem Exam,

Date: Feb 28, 2026

Modern Cryptography

Marks: 20

Instructor: Somitra Sanadhya

Notes: There are 5 questions, each for 4 marks. Marks for each question/part are shown in the right margin. Write briefly. Unnecessarily verbose answers will fetch lower marks.

1. Consider the following encryption scheme: $M = C = K = \{0, 1, 2, \dots, 9\}$,
 $c = \text{Enc}(k, m) = (k + m) \bmod 10$ and $\text{Dec}(k, c) = (c - k) \bmod 10$ (4)

The key generation algorithm chooses a key $k \in K$ uniformly at random from K . Is this scheme perfectly secure? Prove your answer formally.

2. Let encryption of a message m with secret key k by DES be denoted as $c = \text{DES}_k(m)$. Since it is well-known that DES is weak due to its small key-size, your friend suggests increasing the key-size in the following 2 different ways. She suggests picking any one of them as a replacement for the DES.

First, generate two keys k_1 and k_2 and then encrypt the message m as:

(i) $c = \text{DES}_{k_1}(k_2 \oplus m)$

(ii) $c = k_2 \oplus \text{DES}_{k_1}(m)$

For these two encryptions schemes, show that

- (a) Scheme (i) is insecure. (2)
(b) Scheme (ii) is also insecure. (2)

Note: You need to show an attack better than brute-forcing the key for each of these two cases. Assume a suitable attack model and mention it.

3. Show that the Feistel structure is its own inverse for any number of rounds. The only change needed in the decryption operation is to reverse the sequence of the round-keys used during the encryption process. (4)
4. In many instances while using a block-cipher E , we get messages of length that are not multiples of the block-size. Padding is used in these situations. However, this increases the length of the ciphertext.

A mode called "ciphertext stealing" has been suggested to ensure that the ciphertext length is exactly the same as the plaintext length when there is no IV. If the scheme uses an IV, then the ciphertext length is equal to the plaintext length plus the length of the IV.¹ This mode can be used with other standard modes. Figure 1 on the next page shows it used with the ECB mode. The first $(n-2)$ blocks are not shown, but are computed as $C_i = E_k(P_i)$, where E_k denotes the encryption function instantiated with key k .

¹This mode requires messages of length strictly more than 1 block.

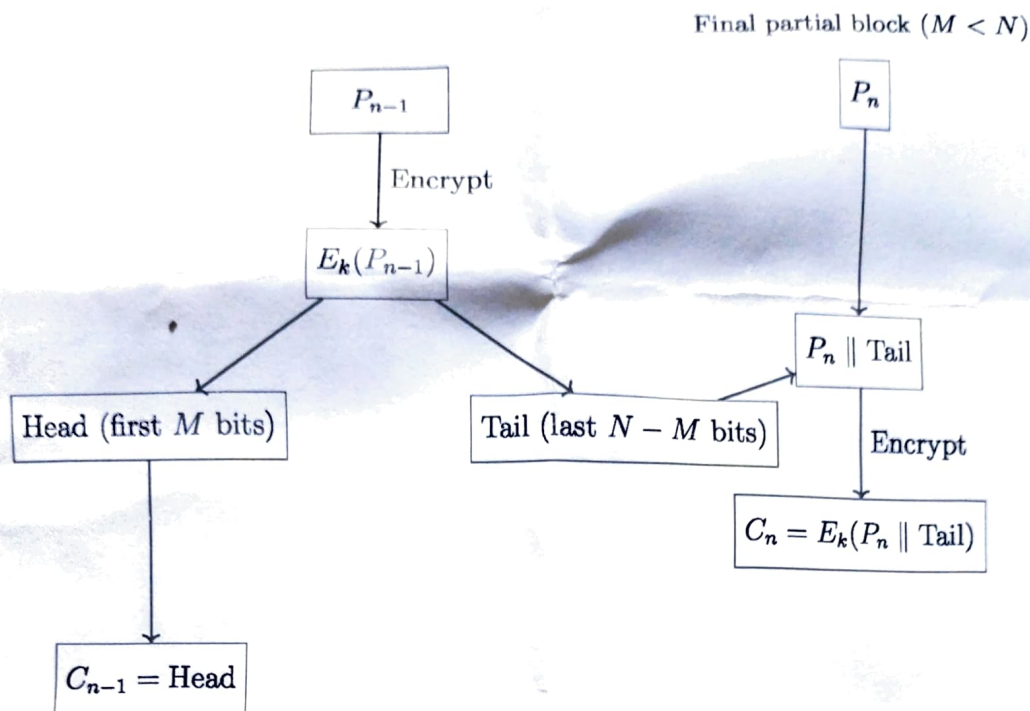


Figure 1: Ciphertext stealing used with the ECB mode. Encryption of last two message blocks is shown.

The last two blocks of the message P_{n-1} and P_n are shown to be encrypted to C_{n-1} and C_n . The block-length of E is N bits and the last block P_n is M bits long. Note that $M < N$.

- (a) Explain the decryption algorithm for ciphertext stealing used with ECB mode, as shown in Figure 1. (1)
- (b) Write the encryption and decryption functions for ciphertext stealing when used with the CBC mode. (3)

5. Answer the following questions in brief.

- (a) What is the key-whitening step and why is it used in the AES? (1)
- (b) A finite field $GF(2^8)$ is used in the AES design. We use hexadecimal notation for this question. The finite field of AES is realized using an irreducible polynomial denoted by 11B. What will be the product of two polynomials 05 and 6E in this field? (1)
- (c) What is the role of an IV in a stream cipher? (1)
- (d) Using a single LFSR to build a stream cipher is not a good idea. Why? (1)

Handwritten notes on the left margin:

$$C_{n-1} = E_k(P_{n-1} \oplus P_n || C_{n-1} [n-2 \dots N-1])$$

$$P_n = P_n || C_{n-1} [0 \dots M-1]$$

$$C_n = E_k(P_n || C_{n-1} [0 \dots M-1])$$

Handwritten note: 1000

Handwritten note: 10 → A

Handwritten note: 1000 8

Handwritten note: 1000 9

Handwritten note: 1010 A

Handwritten note: 1000 1011 B 120

Handwritten note: 0101

Handwritten note: 1110